Stereo calibration targets - OpenCV

For the Garuda bench: Arducam B0267 quad Camarray (OV9281 x2 active) on Raspberry Pi 5

PRINTING

1. Print pages 2-4 at 100% scale ('Actual size' - never 'Fit to page'). Laser printer preferred, matte paper.
2. Verify the 100 mm ruler on each pattern page with a physical ruler before use. If it measures 99 or 101 mm, the printer scaled it: fix the print dialog, or measure the true square size and use THAT value in code.
3. Mount dead flat: spray-glue or tape all edges to glass, acrylic, aluminium sheet or stiff foam board. Any bow or ripple becomes systematic depth error. Rigid is everything.

WHICH TARGET

Page 2 - Checkerboard 9x6 inner corners, 20 mm squares. The OpenCV classic. Whole board must be visible in BOTH cameras in every accepted frame. Best accuracy per frame; use for the primary calibration.

Page 3 - ChArUco 7x5 squares, 24 mm / 18 mm markers, DICT_5X5_50. Tolerates partial views and works at frame edges - useful for covering the corners of the FoV where distortion lives. Optional but recommended.

Page 4 - A second checkerboard at 15 mm squares (11x8 inner corners) for close-range work if the 20 mm board overfills the view at your chosen baseline.

CAPTURE PROCEDURE (B0267: one merged 2560x800 frame -> split into L/R 1280x800 before use)

- Lock exposure and gain (the Camarray shares them across sensors - set once, leave alone).
- Capture 30-50 stereo pairs: near/far, left/right/top/bottom of frame, and TILTED (~30-45 deg skew in several frames - pure frontal views make focal length poorly conditioned).
- Keep the board still per shot (GS kills motion blur, but exposure still integrates).
- Target RMS reprojection error: < 0.3 px per camera, < 0.5 px stereo. Re-shoot outliers, don't average them in.

OPENCV PARAMETERS (page 2 board)

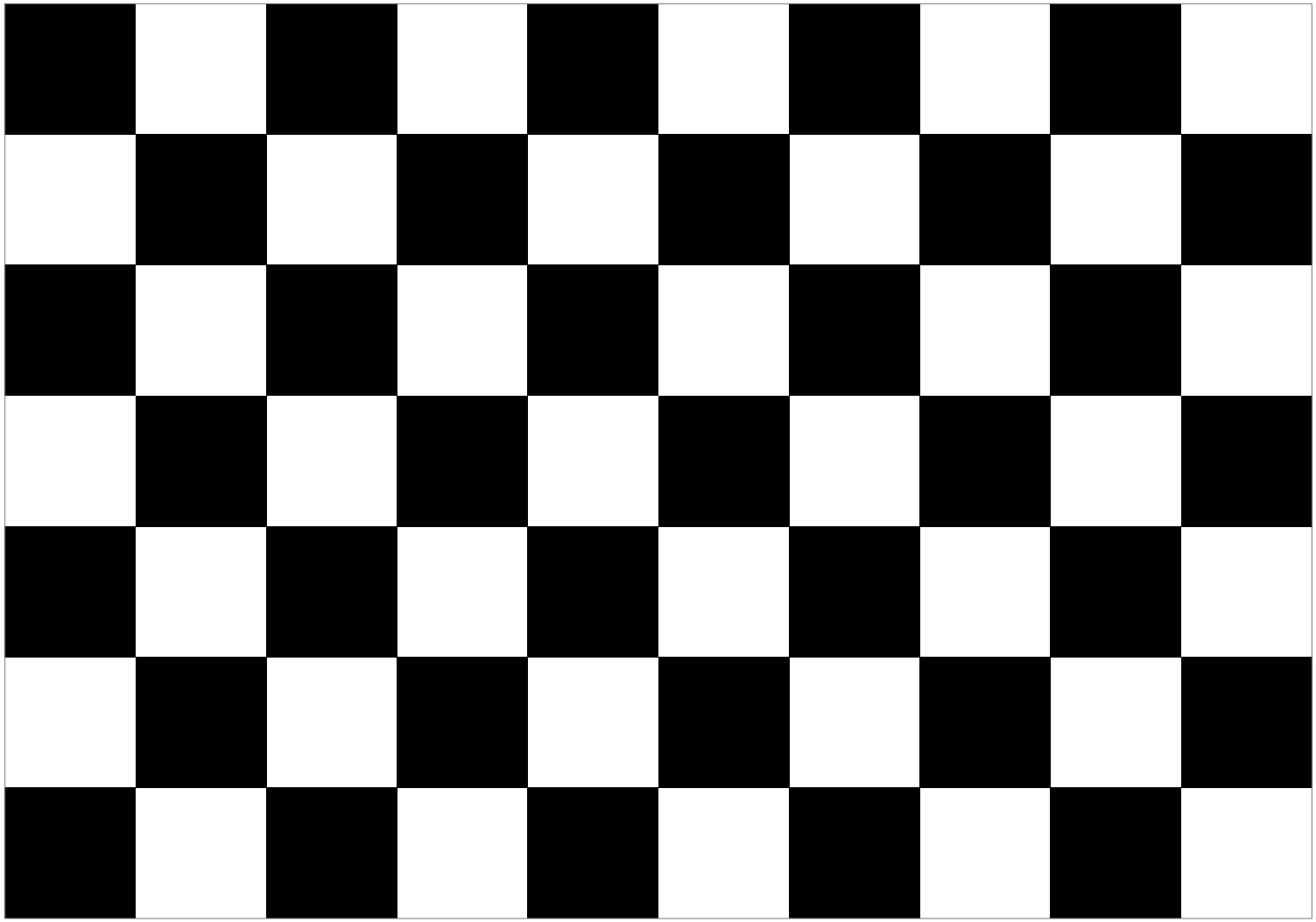
```
pattern_size = (9, 6)          # INNER corners, not squares
square_size  = 20.0            # mm - or your measured value
findChessboardCornersSB(gray, pattern_size)   # SB variant: more robust than the classic finder
stereoCalibrate(..., flags=cv2.CALIB_FIX_ASPECT_RATIO)
then stereoRectify -> initUndistortRectifyMap -> remap, and StereoSGBM on the rectified pair.
```

BOOKKEEPING (CLAUDE.md rules apply)

- Record measured square size, RMS errors, and the calibration YAML into bench/ and commit.
- Calibration is per-rig: redo after ANY change to lenses, baseline, or mounting torque.

Checkerboard - 9 x 6 inner corners - 20 mm squares

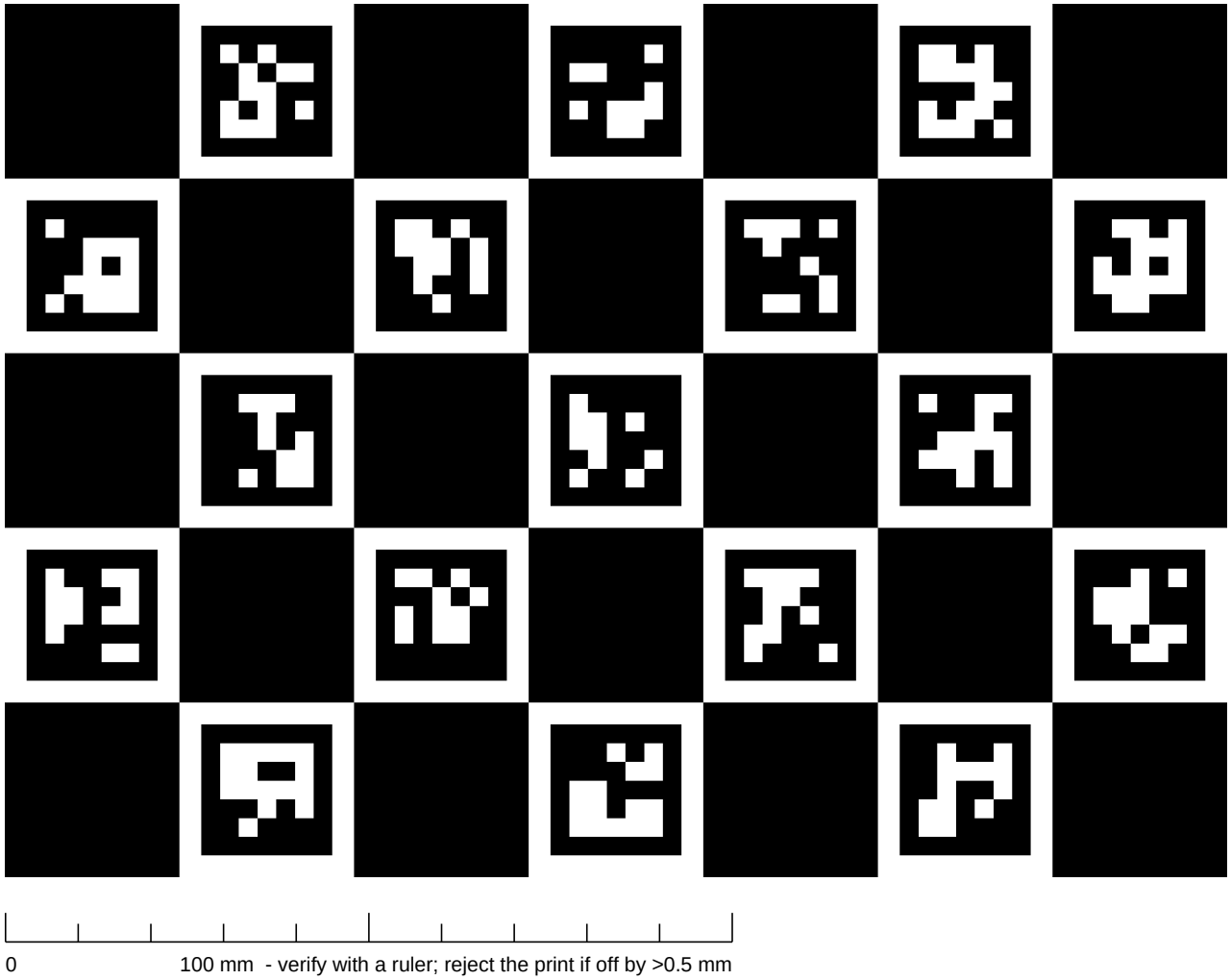
pattern_size=(9,6) square_size=20 mm - keep fully visible in both cameras



0 100 mm - verify with a ruler; reject the print if off by >0.5 mm

ChArUco - 7 x 5 squares - 24 mm squares, 18 mm markers - DICT_5X5_50

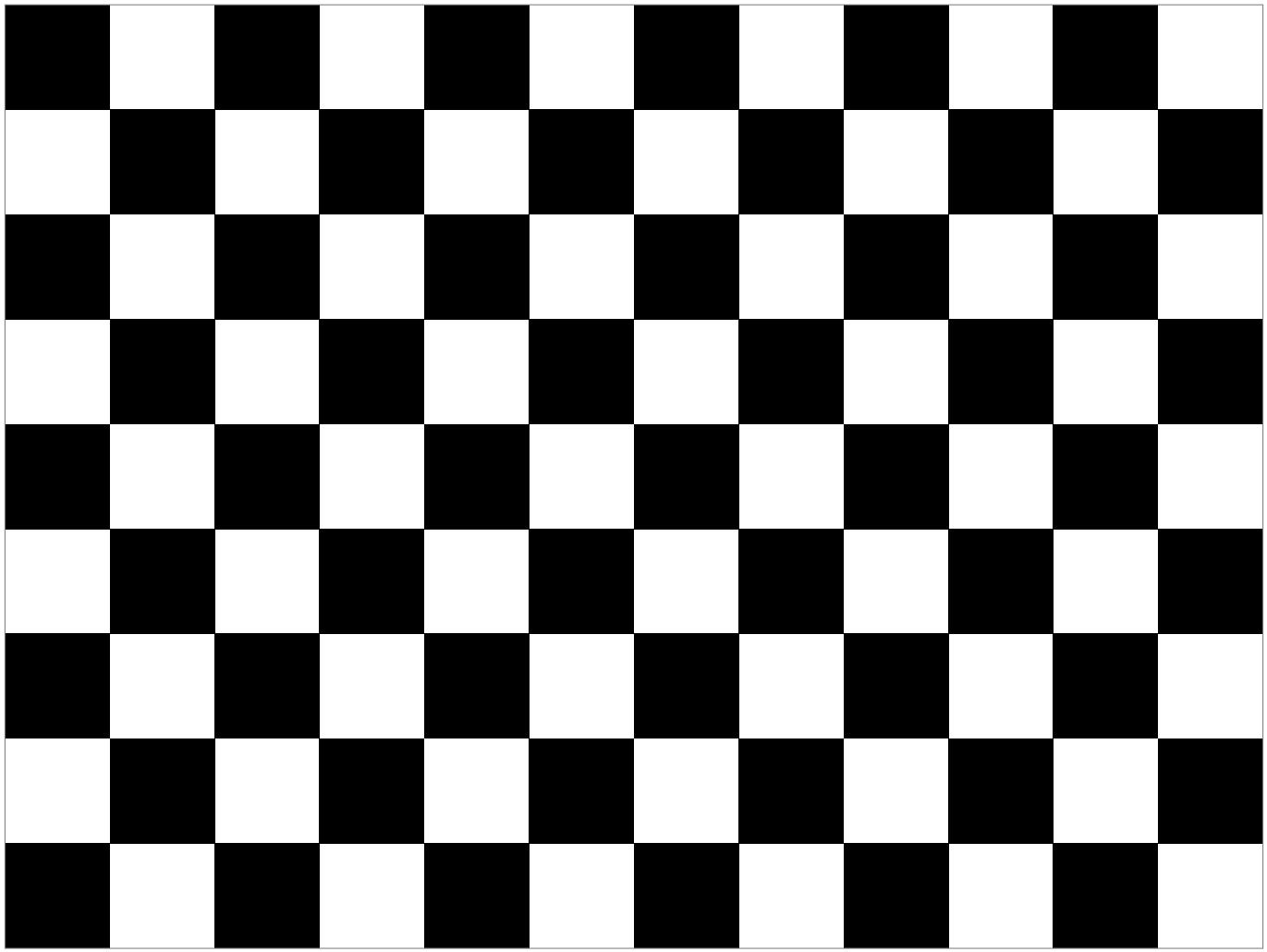
CharucoBoard((7,5), 0.024, 0.018, DICT_5X5_50) - partial views allowed; good for FoV corners



0 100 mm - verify with a ruler; reject the print if off by >0.5 mm

Checkerboard (close-range) - 11 x 8 inner corners - 15 mm squares

pattern_size=(11,8) square_size=15 mm



0 100 mm - verify with a ruler; reject the print if off by >0.5 mm